UNIVERSAL INJECTIVITY OF THE MAP INTO THE ADIC COMPLETION FOR A FINITE FLAT MODULE, UNDER A CONDITION ON THE ASSOCIATED PRIMES OF THE FIBRES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Universal injectivity into the completion from a condition on associated primes of tensor products with field quotients. Same chapter flat.tex, label lemma-universally-injective-to-completion (invoked at proof line 2123). It is the general criterion, whose hypothesis is verified here in the restricted form stated.
- The Bourbaki lemma and remark controlling the associated primes of tensor products by those of the fibres. Algebra, labels algebra-lemma-bourbaki-fibres and algebra-remark-bourbaki (invoked at proof lines 2126 and 2127). They guarantee that the associated primes of the tensor products $N \otimes_R Q$ are contained in the set of associated primes of the modules $N \otimes_R \kappa(\mathfrak p)$, so the hypothesis of the criterion is met.

Source reference. Original lemma 05C2: sources/stacks-chow/tags.txt, source lines 6905–6905. Navigation pointer for \ref{algebra-lemma-bourbaki-fibres} used inside the counted span (line 2126); the argument that consumes it is retained in the span.

Source reference. Original remark 05E0: sources/stacks-chow/tags.txt, source lines 6973–6973. Navigation pointer for \ref{algebra-remark-bourbaki} used inside the counted span (line 2127); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DL: sources/stacks-flat/pinned-original-source.tex, source lines 2060–2060. Navigation pointer for \ref{lemma-universally-injective-to-completion} used inside the counted span (line 2123); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let $R \to S$ be a ring map, and N an S-module. Assume*

(1) *R is a Noetherian ring,*
(2) *S is a Noetherian ring,*
(3) *N is a finite S-module,*
(4) *N is flat over R, and*
(5) *for any prime $\mathfrak{q} \subset S$ which is an associated prime of $N \otimes_R \kappa(\mathfrak{p})$ where $\mathfrak{p} = R \cap \mathfrak{q}$ we have $IS + \mathfrak{q} \neq S$.*

Then the map $N \to N^\wedge$ of N into the I-adic completion of N is universally injective as a map of R-modules.

Proof. This follows from Lemma 05DL because Algebra, Lemma 05C2 and Remark 05E0 guarantee that the set of associated primes of tensor products $N \otimes_R Q$ are contained in the set of associated primes of the modules $N \otimes_R \kappa(\mathfrak{p})$. □